CSE400 – Fundamentals of Probability in Computing
Lecture 19: N-Variate Joint PDF/PMF/CDF

1. Joint Distribution of N Random Variables

1.1 Representation of Multiple Random Variables

Definition

For random variables $X_1, X_2, \ldots, X_N$, the vector form is:

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

The joint distribution describes the probability behavior of all variables together.

It can be represented using:

Joint PMF

Joint PDF

Joint CDF

2. Jointly Gaussian Random Variables

2.1 Joint Gaussian PDF (Two Variables)

For two variables X and Y:

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right)\right.\right.$$

Where:

μ_X, μ_Y: means

σ_X, σ_Y: standard deviations

ρ: Pearson correlation coefficient

3. Example 1: Jointly Gaussian Variables

Given

$$\mu_X = 2, \quad \sigma_X^2 = 1$$

$$\mu_Y = -3, \quad \sigma_Y^2 = 4$$

$$cov(X, Y) = -1$$

(a) Joint PDF Construction

Step 1: Correlation Coefficient

$$\rho_{XY} = \frac{cov(X, Y)}{\sigma_X \sigma_Y}$$

$$\sigma_X = 1, \quad \sigma_Y = 2$$

$$\rho_{XY} = -\frac{1}{2}$$

Step 2: Substitute into General Formula

$$\mu_X = 2,\ \sigma_X = 1,\ \mu_Y = -3,\ \sigma_Y = 2,\ \rho = -0.5$$

Final Joint PDF

$$f_{XY}(x, y) = \frac{1}{2\pi\sqrt{3}} \exp\left(-\frac{2}{3}\left[(x-2)^2 + (x-2)(y+3) + \frac{(y+3)^2}{4}\right]\right)$$

(b) Marginal PDFs

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx$$

Key Property

If X and Y are jointly Gaussian:

$$X \sim N(\mu_X, \sigma_X^2), \quad Y \sim N(\mu_Y, \sigma_Y^2)$$

Marginal Results

$$f_X(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x-2)^2}{2}\right)$$

$$f_Y(y) = \frac{1}{2\sqrt{2\pi}} \exp\left(-\frac{(y+3)^2}{8}\right)$$

(c) Probability $Pr(X < 0)$

Step 1: Identify CDF

$$F_X(x) = Pr(X \leq x), \quad Pr(X < 0) = F_X(0)$$

Step 2: Standardization

$$Z = \frac{X - \mu_X}{\sigma_X}$$

$$Pr(X < 0) = Pr(Z < -2)$$

Step 3: Symmetry Property

$$\phi(-x) = 1 - \phi(x) = Pr(Z > x)$$

$$Pr(Z < -2) = Pr(Z > 2)$$

Step 4: Q-Function

$$Q(x) = Pr(Z > x)$$

Final Answer

$$Pr(X < 0) = Q(2)$$

4. Application: Large MIMO System

System Setup

M: transmit antennas

N: receive antennas

Channel Coefficients

Each path between antenna i and j:

h_{ij}

Channel Matrix

$$H = \begin{bmatrix} h_{11} & h_{12} & \cdots & h_{1N} \\ h_{21} & h_{22} & \cdots & h_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ h_{M1} & h_{M2} & \cdots & h_{MN} \end{bmatrix}_{M\times N}$$

5. Joint PMF

For discrete random variables:

$$P_{X_1,\ldots,X_N}(x_1,\ldots,x_N) = P(X_1 = x_1, \ldots, X_N = x_N)$$

6. Joint CDF

$$F_{X_1,\ldots,X_N}(x_1,\ldots,x_N) = P(X_1 \le x_1, \ldots, X_N \le x_N)$$

7. Joint PDF

$$f_{X_1,\ldots,X_N}(x_1,\ldots,x_N) = \frac{\partial^N}{\partial x_1 \cdots \partial x_N} F_{X_1,\ldots,X_N}(x_1,\ldots,x_N)$$

8. Independence of Random Variables

Condition

$$X = [X_1, X_2, \ldots, X_N]^T$$

If independent:

$$f_X(x_1,\ldots,x_N) = \prod_{i=1}^{N} f_{X_i}(x_i)$$

9. Example 2: Uniformly Distributed 3D Vector

Given

$$|X_1| \le 1,\ |X_2| \le 1,\ |X_3| \le 1$$

$$f_{X_1,X_2,X_3}(x_1,x_2,x_3) = \begin{cases} C, & \text{inside cube} \\ 0, & \text{otherwise} \end{cases}$$

(a) Find C

$$C(2)(2)(2) = 1 \Rightarrow C = \frac{1}{8}$$

(b) Marginal

$$f_{X_1,X_2}(x_1, x_2) = \int_{-1}^{1} \frac{1}{8} dx_3 = \frac{1}{4}$$

(c) Marginal

$$f_{X_1}(x_1) = \int_{-1}^{1} \int_{-1}^{1} \frac{1}{8} dx_2 dx_3 = \frac{1}{2}$$

(d) Conditional PDF

$$f_{X_1|X_2,X_3} = \frac{1/8}{1/4} = \frac{1}{2}$$

(e) Conditional PDF

$$f_{X_1,X_2|X_3} = \frac{1/8}{1/2} = \frac{1}{4}$$

(f) Independence Check

$$LHS = \frac{1}{8}$$

$$RHS = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$$

Conclusion

LHS = RHS $\Rightarrow$ independent

10. Example 3: Trivariate Random Variables

Given

$$f_{XYZ}(x, y, z) = \begin{cases} Ke^{-(ax+by+cz)}, & x, y, z > 0 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find K

$$K \left(\frac{1}{a}\right) \left(\frac{1}{b}\right) \left(\frac{1}{c}\right) = 1$$

$$K = abc$$

(b) Marginal Joint PDF

$$f_{XY}(x, y) = \int_0^\infty abce^{-(ax+by+cz)} dz = abe^{-(ax+by)}$$

(c) Marginal PDF

$$f_X(x) = \int_0^\infty abe^{-(ax+by)} dy = ae^{-ax}$$

(d) Independence Check

LHS $= abce^{-(ax+by+cz)}$

RHS $= (ae^{-ax})(be^{-by})(ce^{-cz})$

Conclusion

LHS = RHS $\Rightarrow X, Y, Z$ are independent